

Week 3: CSS Notes

1. CSS3 Essentials

What is CSS3?

- CSS3 is the latest version of CSS, introducing powerful features such as Flexbox, animations, transforms, and custom properties.
- **Why It Matters:** Enables dynamic, responsive designs, such as those found on e-commerce sites (e.g., Myntra's clean layouts).
- **Key Features:** Flexbox, Grid, transitions, calc(), and pseudo-elements (::before, ::after).

Browser Compatibility & Vendor Prefixes

- **Vendor Prefixes:** Ensure cross-browser support for experimental features.
 - Examples: -webkit-transition, -moz-transform.

Example: Smooth hover effect.

```
.button {  
  transition: all 0.3s ease;  
  -webkit-transition: all 0.3s ease;  
  }
```

- **Tip:** Use Can I Use to check compatibility. Modern browsers often don't need prefixes.
- **Troubleshooting:** If a property doesn't work, check browser support or add prefixes via tools like Autoprefixer.

CSS3 Features

- **Calc Function:** Dynamically calculate values.

Example: Responsive width.

```
.container {  
  width: calc(100% - 2rem);  
  }
```

- **Pseudo-elements (::before, ::after):** Add decorative content.

Example: Add a decorative arrow.

```
.cta-button::after {  
  content: "→";  
}
```

margin-left: 5px;

- }
- **CSS Variables:** Reuse values for consistency.

Example: Define brand colors.

```
:root {  
  --primary: #3498db;  
  --accent: #2ecc71;  
}  
.header {  
  background: var(--primary);  
  
  ○ }
```

2. CSS Flexbox

- **Purpose:** Simplifies layouts for rows or columns, perfect for headers and grids.
- **Key Properties:**
 - **display: flex:** Turns a container into a flexbox.
 - **justify-content: center | space-between:** Aligns items horizontally.
 - **align-items: center:** Aligns items vertically.
 - **flex-wrap: wrap:** Allows items to wrap for responsiveness.
 - **gap: 1rem:** Adds spacing between items.

Example: Header with logo and menu (like Myntra's navbar).

```
#header {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  padding: 1rem;  
  background: var(--primary);  
  
  ● }
```

- **Troubleshooting:**
 - **Items not aligning?** Check flex-direction (default is row).
 - **Items shrinking?** Use flex-shrink: 0 or min-width.

3. CSS Positioning

- **Types:**

Fixed: Stays fixed on the screen (e.g., a "Contact Us" button).

```
.contact-btn {  
  position: fixed;
```

```
bottom: 1rem;  
right: 1rem;
```

```
    }  
}
```

Sticky: Sticks when scrolling (e.g., navbar).

```
#header {  
  position: sticky;  
  top: 0;  
  z-index: 1000;
```

```
    }  
}
```

Relative: Moves relative to its normal position.

```
.card {  
  position: relative;  
  top: 10px;
```

```
    }  
}
```

Absolute: Positions relative to the nearest positioned ancestor.

```
.dropdown {  
  position: absolute;  
  top: 100%;  
  left: 0;
```

```
    }  
}
```

- **Display Properties:**

- **display: none:** Hides elements (e.g., dropdowns).
- **display: block | flex:** Shows elements on interaction.

- **Z-index:** Controls stacking order.

Example: Keep dropdowns on top.

```
.dropdown {  
  z-index: 1000;
```

```
    }  
}
```

- **Troubleshooting:**

- **Z-index not working?** Ensure the element has position: relative | absolute | fixed.
- **Absolute element misplaced?** Add position: relative to its parent.

4. Cursor & Scroll

- **Cursor:** Enhances interactivity.

Example: cursor: pointer for buttons.

```
.cta-button {
  cursor: pointer;
```

```
    }
}
```

- **Scroll:**

- overflow-y: auto: Adds a vertical scrollbar when needed.
- scroll-behavior: smooth: Smooth scrolling for links.

Example: Scrollable services section.

```
#services {
  max-height: 400px;
  overflow-y: auto;
  scroll-behavior: smooth;
```

```
    }
}
```

- **Troubleshooting:** If scrolling isn't smooth, ensure html { scroll-behavior: smooth; }.

5. CSS Transforms

- **Purpose:** Alter elements' appearance (move, rotate, scale, skew).
- **2D Transforms:**
 - **Scale:** Resize elements.

Example: Zoom image on hover (like Myntra's product cards).

```
.card img:hover {
  transform: scale(1.1);
  transition: transform 0.3s ease;
```

```
    }
}
```

- **Rotate:** Spin elements.

Example: Rotate icon on hover.

```
.icon:hover {
  transform: rotate(360deg);
  transition: transform 0.5s ease;
```

```
    }
}
```

- **Translate:** Move elements.

Example: Slide-in dropdown.

```
.dropdown {
  transform: translateY(10px);
  opacity: 0;
  transition: all 0.3s ease;
}
```

```
.dropdown:hover {
  transform: translateY(0);
  opacity: 1;
```

```
  }
}
```

- **Skew:** Slant elements.

Example: Decorative banner.

```
.banner {
  transform: skew(-15deg);
```

```
  }
}
```

- **3D Transforms:**
 - Use perspective for 3D effects.

Example: Flip box (like a product card flip).

```
.flip-box {
  perspective: 1000px;
}
.flip-box-inner {
  transition: transform 0.6s;
  transform-style: preserve-3d;
}
.flip-box:hover .flip-box-inner {
  transform: rotateY(180deg);
}
.flip-box-front, .flip-box-back {
  position: absolute;
  backface-visibility: hidden;
}
.flip-box-back {
  transform: rotateY(180deg);
```

```
  }
}
```

- **Troubleshooting:** Add transition for smooth transforms. Use transform-origin to control the pivot point.

6. CSS Animations

- **Keyframes:** Define animation steps.
 - **Percentage Division:** For complex animations.

Example: Solar system orbit (like an interactive e-commerce feature).

```
@keyframes orbit {
  0% { transform: rotate(0deg) translateX(100px) rotate(0deg); }
  100% { transform: rotate(360deg) translateX(100px) rotate(-360deg); }
```

```
}
.planet {
  animation: orbit 5s linear infinite;
```

```
  ■ }
```

- **From-To Division:** For simple animations.

Example: Image spin.

```
@keyframes spin {
  from { transform: rotate(0deg); }
  to { transform: rotate(360deg); }
}
.image:hover {
  animation: spin 1s linear;
```

```
  ■ }
```

- **Background Color Animation:**

Example: Color-changing section.

```
@keyframes bgChange {
  0% { background: var(--primary); }
  50% { background: var(--accent); }
  100% { background: var(--primary); }
}
.section {
  animation: bgChange 5s infinite;
```

```
  ○ }
```

- **Wave Effect:**

Example: Animated wave (like Myntra's footer effects).

```
@keyframes wave {
  0% { transform: translateX(0); }
  100% { transform: translateX(-50%); }
}
.wave {
  background: url('data:image/svg+xml;utf8,<svg viewBox="0 0 1200 120"
xmlns="http://www.w3.org/2000/svg"><path
d="M0,0V46.29c47.79,22.2,103.59,32.17,158,28,70.36-5.37,136.33-33.31,206.8-37.5C438.
64,32.43,512.34,53.67,583,72.05c69.27,18,138.3,24.88,209.4,13.08,36.15-6,69.85-17.84,10
4.45-29.34C989.49,25,1113.64,31.08,1200,41.05V0Z" fill="white"/></svg>');
  animation: wave 7s linear infinite;
```

```
  ○ }
```

- **Bubble Animation:**

Example: Rising bubbles for background flair.

```
@keyframes bubble {
  0% { transform: translateY(0); opacity: 0.5; }
  100% { transform: translateY(-100vh); opacity: 0; }
}

.bubble {
  animation: bubble 10s linear infinite;
```

- }
- **Troubleshooting:**
 - **Animation not running?** Check animation syntax: name duration timing-function.
 - **Choppy animation?** Use will-change: transform for better performance.

7. Dropdown & Mega Menu

- **Dropdown Menu:**
 - Use position: absolute and transform: translateY for smooth entry.

Example: Simple dropdown (like Myntra's category menu).

```
.nav-menu ul {
  display: none;
  position: absolute;
  transform: translateY(10px);
  opacity: 0;
  transition: all 0.3s ease;
}

.nav-menu:hover ul {
  display: block;
  transform: translateY(0);
  opacity: 1;
```

- }
- **Mega Menu:**
 - Use display: flex for multi-column layout and transform: scale for zoom.

Example: Services mega menu.

```
.mega-menu-content {
  display: none;
  position: absolute;
  display: flex;
  transform: scale(0.8);
  opacity: 0;
  transition: all 0.3s ease;
}
```

```
.mega-menu:hover .mega-menu-content {
  display: flex;
  transform: scale(1);
  opacity: 1;
  ○ }
```

- **Troubleshooting:** If menus overlap, adjust z-index. Ensure parent has position: relative.

8. Opacity & Z-index

- **Opacity:** Controls transparency (0 to 1).

Example: Fade-in dropdown.

```
.dropdown {
  opacity: 0;
  transition: opacity 0.3s ease;
}
.dropdown:hover {
  opacity: 1;
  ○ }
```

- **Z-index:** Manages stacking.

Example: Keep navbar above content.

```
#header {
  z-index: 1000;
  ○ }
```

- **Troubleshooting:** If an element is hidden behind others, increase its z-index and ensure it's positioned.

9. Practical Examples

- **Flip Box:** 3D card flip for service details.
- **Solar System:** Orbiting planet for interactive visuals.
- **Wave Effect:** Animated footer wave for style.
- **Bubble Animation:** Background bubbles for dynamic flair.
- **Image Effects:** Scale, rotate, and translate on hover for product-like cards.